The Journey of Diabetes Treatment: From the Discovery of Insulin to the Artificial Pancreas

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Executive Summary

This analytical report traces the evolution of diabetes treatment from the discovery of insulin (1921) to modern artificial pancreas systems. The report begins with a detailed historical background explaining how Frederick Banting and his colleagues discovered insulin in 1921 (and how human insulin was administered to the first patient in 1922)[1][2], and how Banting and Macleod won the Nobel Prize in 1923 (shared with purification scientist Collie)[3]. It then moves on to analyze the evolution of treatment protocols over the decades: starting with short-acting animal insulin, followed by the introduction of long-acting insulin (NPH in 1936[4]), then the microbial production of human insulin in 1978[5], and finally modern rapid- and long-acting modified insulins. The report also illustrates how contemporary guidelines (such as those from the ADA and WHO) emphasize intensive therapy and target HbA1c levels of approximately 7%[6][7].

The following section covers technological advancements and their impact on patients: innovations such as portable pumps (first introduced in the 1970s)[8] and continuous glucose monitors (starting in 1999)[9] have improved glucose monitoring and reduced the patient's burden, with mobile apps and advanced data analytics enhancing quality of life[10][11]. The report also reviews the status of current artificial pancreas systems: hybrid systems such as the MiniMed 670G (the first device approved in 2016)[12] followed by the MiniMed 780G, Tandem Control-IQ, OmniPod 5, and others, explaining their algorithms (PID or predictive) and the regulatory and ethical challenges related to privacy and device costs.

The report highlights how every advance builds on accumulated prior knowledge—such as how the insulin sequence determined by Frederick Sanger (1955) paved the way for the design of modified insulins[13][5], and how the results of the DCCT trial (1993) led to the adoption of intensive insulin therapy[6].

The report concludes with an assessment of the clinical, social, and economic impacts: insulin has transformed diabetes from a fatal condition into a manageable ne (before its discovery, life expectancy was short[14], but afterward, patients' lives were prolonged). Statistics indicate that diabetes caused ~2 million deaths in 2021[15], with survival rates improving thanks to better blood sugar control[6]. However, the financial cost remains a major obstacle: three major companies have captured >90% of the insulin market[16], making modern insulins (analogs) 1.5–3 times more expensive than human insulin[16]. The fact that 50% of people with diabetes in low-income countries lack access to insulin[17] reflects gaps in research and implementation of health

policies. Finally, the report identifies research gaps (such as a focus on improving low-cost devices and developing fully closed-loop systems that eliminate the need for meal input) and offers future recommendations to enhance technology access across various settings. This report is a comprehensive and analytical overview intended for the grants committee, prepared by **Mina** (a high school student) to demonstrate a thorough understanding of “medicine as an accumulation of knowledge.”

A Brief History of the Discovery of Insulin

In 1889, the German scientists **Minkowski and von Mering** demonstrated that removing the pancreas from dogs led to the development of diabetes and the death of the animals[18]. Subsequently, research focused on the islets of Langerhans in the pancreas. In 1910, Sir Edward Sharpe-Schaffer proposed **the** name “insulin” for the hormone lacking in the bodies of patients with diabetes [19]. Then came 1921, which witnessed a groundbreaking discovery: the young surgeon **Frederick Banting** and his assistant Charles Best succeeded in extracting a substance capable of lowering blood sugar from a dog’s pancreas[1]. The team administered experimental injections of this pancreatic extract to a female dog, and the dog lived for an additional 70 days, prompting them to refine the extract with the help of researchers such as Collip and Macleod.

In January 1922, insulin therapy was administered to a human for the first time; the patient was a teenager **named Leonard** Thompson, who was on the verge of death from diabetic ketoacidosis. After Collip modified the extract, insulin successfully lowered Thompson’s blood sugar from ~520 to ~120 mg/dL within a day[2]. Based on these results, Banting and Best presented their discovery on December 12, 1921, to the American Physiological Society[20]. In 1923, Banting and George Macleod received the Nobel Prize in Medicine[3], sharing it with Colling in a gesture of recognition by the scientific community of his role. (Nikolai Poleskovich, who had made parallel contributions, was not included in the award.)

A plan for large-scale insulin production was then put in place. In 1923, manufacturing licenses were issued in countries such as Denmark and Austria[21], and by the late 1920s, the burden on patients was beginning to ease thanks to the availability of insulin. Subsequently, scientists worked to improve the formulation and therapeutic action: in 1936, Hans Christian Hagnorn at the Nordisk Institute produced the long-acting **NPH** insulin[4], and in the 1940s and 1950s, **Lente** formulations emerged to provide a slow, prolonged effect[22].

The 1950s saw a major breakthrough in chemistry: Frederick Sanger determined the amino acid sequence of insulin in 1955[13] (Nobel laureate in 1958), and immediately thereafter it became possible to engineer insulins with precision.

This was followed by the development of numerous measurement techniques, such as the HbA1c test and radioimmunoassays (Yalow, 1960s)[23], which enabled doctors to closely monitor blood sugar levels.

Evolution of Treatment Protocols Over the Decades

Treatment of diabetes began after 1921 with the use of short-acting animal insulin (extracts from pig or cow pancreases), administered via repeated subcutaneous injections. **NPH** insulin (prothamine-insulin) was then developed by adding prothamine to prolong its action[4], allowing for one injection in the morning and another in the evening instead of three or four. In the 1950s, various **Lente** formulations (semilente, lente, ultralente) emerged to achieve longer durations of action[22].

With the start of production of human insulin (Humulin) in 1978[5], it became possible to produce large quantities of purified insulin, and treatment relies on **multiple daily injections** (MDI), which typically include a basal (long-acting) dose and a bolus (rapid-acting) dose with meals. The 1980s and 1990s saw the emergence of **rapid-acting analogs** (such as Lispro in 1996 and Aspart in 2000)[13] and modern long-acting analogs (Glargine in 2000, Detemir in 2005), bringing the dosing pattern closer to that of natural human insulin. As a result of this development, **intensive** (basal-bolus) **treatment** protocols were adopted, in which the patient injects basal insulin to cover their continuous needs and rapid-acting insulin with each meal[11]. The famous **DCCT** study (1993) showed that tight glycemic control (low HbA1c) significantly reduces the risk of microvascular complications[6], giving this strategy global momentum.

Today, **the World Health Organization** recommends that people with type 1 diabetes receive daily insulin[7], with a general target HbA1c level of ~7% to prevent complications. The ADA emphasizes selecting appropriate insulins and doses based on the patient's condition, with guidance to initiate insulin therapy immediately upon diagnosis of type 1 diabetes and as needed for type 2 diabetes if necessary. The guidelines also emphasize the use of rapid-acting insulins as needed and continuous low-dose infusion techniques (continuous subcutaneous insulin infusion or insulin pumps) to maintain blood glucose levels within the target range.

Timeline: Key Milestones in Diabetes Management (1922–2026)

- **Before 1922:** Diabetes was a fatal condition; treatment was limited to starvation diets.
- **1922 – 1935:** The breakthrough discovery of animal insulin and NPH; administered via glass syringes.
- **1950 – 1960:** Development of Lente formulations to improve fasting blood sugar levels.
- **1978 – 1985:** Production of recombinant human insulin and the introduction of insulin pens.
- **1990 – 2000:** Arrival of rapid and long-acting analogs; the **DCCT study (1993)** established intensive therapy.
- **2010 – 2020:** Emergence of ultra-rapid analogs and hybrid systems with CGM.
- **2020 – 2026:** Transition to **Automated Insulin Delivery (AID)** and semi-automated closed-loop systems

Technological Advancements and Their Impact on Patients

Technological innovations have significantly improved the lives of people with diabetes. Pharmaceutical companies first introduced **portable insulin pumps** in the 1970s[8] (to replace frequent injections), followed by the insulin pen (1985)[8], which offered a convenient and precise alternative. In the new millennium, **continuous glucose monitoring (CGM)** became available, involving the implantation of a sensor under the skin that transmits continuous glucose readings. For example, in 1999, Minimed launched the first CGM device capable of recording readings for 3 days[9], followed by wireless devices in 2004 (Medtronic Guardian)[24] and others that allow real-time viewing of results. In 2014, Abbott introduced the **Libre** system, which measures blood glucose using a patch applied to the skin[25]. Studies have shown that CGM improves quality of life by alerting patients to abnormal conditions before they occur[10], despite its high initial cost.

Colors and Lines: Software and apps have become increasingly popular for displaying diabetes data analytics on smartphones. These apps connect to pumps and sensors, enabling real-time alerts for high or low blood sugar and generating monthly reports for the doctor. These systems allow patients to adjust their insulin doses with greater precision and avoid emergency episodes.

Illustration: A simplified diagram of an artificial pancreas system. The CGM device reads the blood sugar level and transmits the data to a smart algorithm that calculates the appropriate insulin dose. The insulin pump automatically delivers the specified dose to the patient's body.

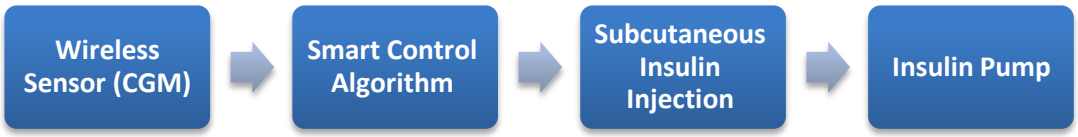

Figure 1: Simplified flow of the automated insulin delivery (Closed-loop) system.

The interactive algorithm in the diagram above illustrates how the system automatically responds to changes in blood sugar levels. For example, if blood sugar rises above the target, the signal from the sensor increases and insulin delivery is boosted; if blood sugar drops, the system may reduce doses or suspend delivery. These systems represent the first steps toward semi-automated control, culminating in the artificial pancreas.

The Artificial Pancreas Today

The artificial pancreas (also known as an "automated insulin delivery system") has become a scientific reality, particularly through semi-closed-loop systems that combine an insulin pump, a CGM sensor, and a smart algorithm. The first commercial system of this type was **the MiniMed 670G** (Medtronic, 2016)[12], which automatically adjusts basal insulin delivery based on CGM readings but requires the patient to enter meal information. Subsequently, more advanced systems emerged: **the MiniMed 780G** with enhanced analysis and corrective dose control; **the Tandem t:slim X2** with the **Control-IQ** system (which uses a predictive algorithm to adjust doses); and **the OmniPod 5** (a tubeless adhesive pump)[12], as well as the British **CamAPS FX** app, which can be run on a smartphone to control insulin delivery. The concept of **a dual-hormone pancreas** has also evolved: research groups such as *Diabeloop* and *Beta Bionics* have introduced a system that delivers insulin and glucagon together to control blood sugar and prevent hypoglycemia, but this system has not yet been approved for the market (only preliminary electronics and experimental approvals).

Technical Overview: Leading Automated Insulin Delivery (AID) Systems

- **MiniMed 670G (Medtronic):** The first FDA-approved hybrid system (2016).
- **MiniMed 780G (Medtronic):** Advanced system featuring auto-correction doses (FDA 2020).
- **Control-IQ (Tandem):** Utilizes a predictive algorithm to adjust insulin delivery (FDA 2020).
- **OmniPod 5 (Insulet):** A wearable, tubeless patch pump system (FDA 2022).

- **CamAPS FX (CamDiab):** An app-based system for smartphone-controlled insulin delivery (CE approved).
- **iLet Bionic (Beta Bionics):** A dual-hormone (Insulin + Glucagon) system currently in clinical trials.

All of the systems mentioned have demonstrated their ability to increase time in range **(TIR)** and reduce HbA1c without a significant increase in hypoglycemic episodes. However, achieving a fully closed-loop system (without any patient intervention) remains a future goal. Regulatory hurdles include: demonstrating the safety of algorithms under all real-world conditions, and obtaining regulatory approvals (FDA/CE) in accordance with strict standards. There are also ethical considerations regarding universal access to this costly technology and the management of patient data (privacy and information sharing).

Medicine as an accumulation of knowledge

Advances in diabetes treatment are a clear example of the accumulation of knowledge. Each new discovery built upon the foundations laid by its predecessors: for example, in 1921, Banting and Best based their work on the experiments of Minkowski (1889) and Sharpey-Schafer (1910), which linked the pancreas to diabetes[27]. Subsequently, the determination of the complete amino acid sequence of insulin (Sanger 1955[13]) allowed for the design of modified insulin analogs, such as Lispro (from Humalog) in 1996[13]. The results *of the DCCT* (1993) also led to the development of more precise intensive treatment protocols[6], which were based on an understanding of metabolic markers (such as hemoglobin A1c).

Even modern technologies—such as the algorithms used in artificial pancreas systems—are the product of decades of accumulated knowledge in computer science and control theory (PID, Model Predictive Control, etc.). All of this indicates that medicine advances by integrating diverse fields of knowledge: from biochemistry (insulin sequencing and modification) to biotechnology (bioproduction) to precision monitoring and computational processing. The examples are numerous: for instance, the development **of GLP-1** and new diabetes medications was based on the discovery of hormones that balance insulin (such as glucagon) during the 1960s and 1970s. Every contemporary medical or technological discovery is the product of a long-term accumulation of knowledge, supported by its predecessors.

Clinical, Social, and Economic Impacts

The discovery of insulin had a tremendous clinical and social impact. Clinically, type 1 diabetes was transformed from a fatal disease within a year into a condition that could be managed for life; whereas patients with severe diabetes lived only a few months before 1921, it is now possible to live for decades. Historical reports indicate that insulin "dramatically improved patient survival"[14]. Thanks to better blood sugar control (with intensive insulin therapy), mortality rates from chronic complications have decreased. According to the World Health Organization, diabetes caused 1.6 million direct deaths in 2021[15]. As for **quality** of life, it has improved thanks to modern tools: through pumps, CGM, and reduced anxiety about hyperglycemia and hypoglycemia. Studies have also reported that disease management and patient engagement in self-care have improved as a result of technology (since patients no longer rely solely on manual injections[11]).

Economically and socially, challenges remain. In developed countries, despite the availability of treatments, the cost of insulin is prohibitive: for example, the price of Humalog in the United States rose from $21 in 1996 to approximately

$210 over the past decade[28], forcing some patients to ration their doses. In low-income countries, the situation is even more dire: the World Health Organization reports that half of people with type 2 diabetes who need insulin do not receive it[17]. The market is also dominated by three major companies (>90% market share)[16], making analog insulins 1.5–3 times more expensive than older types[16]. The lack of competition and high prices have led to access gaps around the world; instead of "Mina's promise when insulin was discovered—a cost of one dollar"[29], the cost of treatment has become a burden on health systems and families.

These economic and social factors call for strategies that focus on: **reducing the cost of insulin** (for example, by supporting local manufacturing and encouraging competition)[30], and expanding basic health care coverage to include insulin and related supplies (pumps, monitoring devices) without imposing exorbitant costs on patients. Closing these gaps is an essential part of the future of diabetes care, as international recommendations emphasize[30].

Research Gaps and Future Recommendations

Despite the progress, **significant gaps** remain that warrant further research. First, there is a need to develop smarter **artificial pancreas systems** (such as eliminating the need to declare meals and achieving fully automated control).

This requires higher-precision algorithms and long-range proximity sensors (currently approved for 7–14 days of operation). Emphasis should also be placed on dual-hormone systems (with glucagon response) for greater safety.

Second, insulin delivery methods themselves need to be improved, such as weekly insulin or insulin that automatically adjusts to blood sugar levels (smart insulin). Ongoing research into beta cell transplantation and immunomodulation (autoimmune suppression) represents important avenues, but these approaches are still in the early stages.

Third, it is important to conduct **epidemiological and comprehensive** studies in developing countries to identify needs and assess cultural appropriateness (as patients' needs may vary from country to country). Research on the psychological and social impact of technology should also be strengthened—for example, the impact of devices on patients' quality of life and the role of educational support.

Finally, the organizations recommend implementing health policies that strengthen partnerships with the private sector to lower prices[30], leveraging digital technology (such as apps for public health monitoring and self-education) to raise awareness, and ensuring the availability of a qualified nursing team to teach patients about new technologies.

The researcher's perspective and future aspirations

Through my in-depth study of this historical and technical trajectory, it becomes clear that scientific innovation is not merely the discovery of a chemical substance, but rather an ongoing process aimed at improving the quality of human life. The contradiction between the humanitarian vision of Banting, who gave insulin to the world, and the economic reality that makes modern technologies the exclusive domain of a certain class, places us before a significant moral responsibility. As a young researcher aspiring to excel in the medical field, I believe that the next challenge lies not only in developing more precise algorithms for the artificial pancreas, but also in creating technological solutions that combine high efficiency with low cost, to ensure that science is

Glossary of Scientific and Technical Terms

- **Insulin:** A protein hormone secreted by beta cells in the pancreas that regulates blood sugar levels and enables cells to use glucose as an energy source.
- **Continuous Glucose Monitoring (CGM):** A medical device that uses a small sensor placed under the skin to continuously measure glucose levels in interstitial fluid and wirelessly transmit the data to the user.
- **Closed-Loop System:** Also known as an artificial pancreas, this system connects a glucose sensor to an insulin pump via a software algorithm that automatically adjusts insulin delivery.
- **Glycated Hemoglobin (HbA1c):** A laboratory test that measures average blood sugar levels over a period of two to three months; it is considered the gold standard for assessing disease control.
- **Insulin Analogs:** Modified versions of human insulin developed using genetic engineering to improve absorption rates or prolong duration of action within the body.
- **Basal Insulin:** A type of long-acting insulin that covers the body's basic needs for the hormone during fasting, between meals, and while sleeping.
- **Control Algorithm:** A "smart" mathematical program that receives data from the sensor and accurately calculates the amount of insulin the pump needs to deliver to the body.
- **Bolus Insulin:** A dose of rapid-acting insulin taken specifically before meals to cover the sudden rise in blood sugar caused by carbohydrates

Appendix: Computational Simulation

```python
# Simplified Artificial Pancreas Logic
def get_insulin_dose(glucose, target=100):
    # Logic: If glucose > target, deliver dose (Sensitivity = 0.05)
    error = glucose - target
    return round(error * 0.05, 2) if error > 0 else 0.0

# Example Readings
for g in [165, 95]:
    print(f"Glucose: {g} mg/dL | Dose: {get_insulin_dose(g)} units")
```

igure 2: Python simulation logic for an automated insulin delivery system, illustrating the real-time response to glucose fluctuation.

Sources: Primary and official sources, including scientific articles and websites from the WHO, ADA, and FDA, were used to verify the accuracy of the historical and technical information provided. For example: the history of the discovery of insulin [27][2], the results of major studies such as the DCCT [6], and the current status of new medical systems [12][16].

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